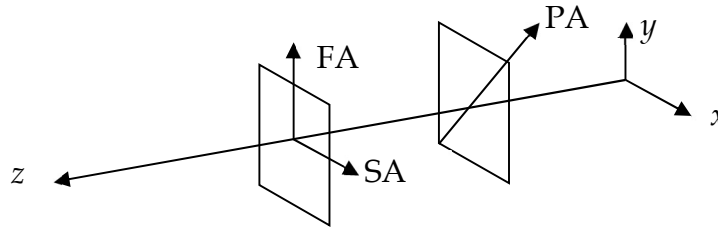
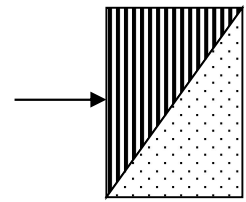


PHL1010: Practice Problems IV
(18/11/2025)

1. An unpolarized beam falls at a linear polarizer (LP) whose pass-axis makes 45° from the horizontal. A Quarter Wave Plate (QWP) is kept in front of this linear polarizer with its fast-axis in the vertical direction as shown in the figure. Write down the electric field after the LP, after the QWP and hence, find out the output state-of-polarization.



2. A Wollaston prism made of calcite is shown in the following figure. If an unpolarized light falls on this prism at a normal incidence, draw the ray propagation through the prism depicting the field vibrations. Determine the angle of deviation between the two emergent beams if the prism angle is 20° and $n_o = 1.658$, $n_e = 1.486$.



3. A left circularly polarized beam ($\lambda_0 = 5893 \text{ \AA}$) is incident normally on a calcite crystal (with its optic axis cut parallel to the surface) of thickness 0.005141 mm . What will be the state of polarization of the emergent beam?
4. Discuss the state of polarization when the x and y components of the electric field are given by the following equations:
- $$E_x = E_0 \cos(kz + \omega t)$$

$$E_y = (E_0/\sqrt{2}) \cos(kz + \omega t + \pi)$$
 - $$E_x = E_0 \sin(kz + \omega t)$$

$$E_y = E_0 \cos(kz + \omega t)$$
 - $$E_x = E_0 \sin(kz - \omega t + \pi/3)$$

$$E_y = E_0 \sin(kz - \omega t - \pi/6)$$
 - $$E_x = E_0 \sin(kz - \omega t + \pi/4)$$

$$E_y = (E_0/\sqrt{2}) \sin(kz - \omega t)$$

In each case, plot the rotation of the tip of the electric vector on the plane $z = 0$.